



What is a Network?



A network is a connection system where people, places, or devices can communicate and exchange something.

Examples:

- Road network connects places.
- Mobile network connects phones.
- Social network connects people.
- Computer network connects digital devices.

Can our classroom be called a network?



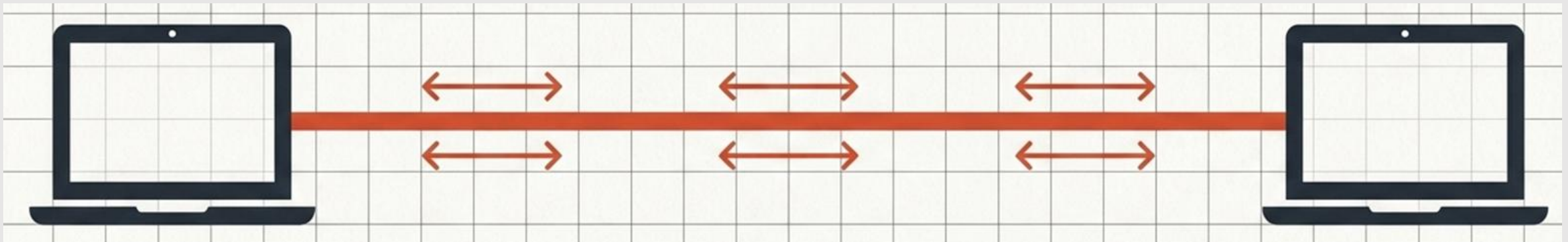
What is a Computer Network?



A computer network is a system where two or more computers or digital devices are connected to share data, hardware, software, and communication services.

Example:

Many computers in a lab can use one printer and one internet connection.





Objectives of Networking



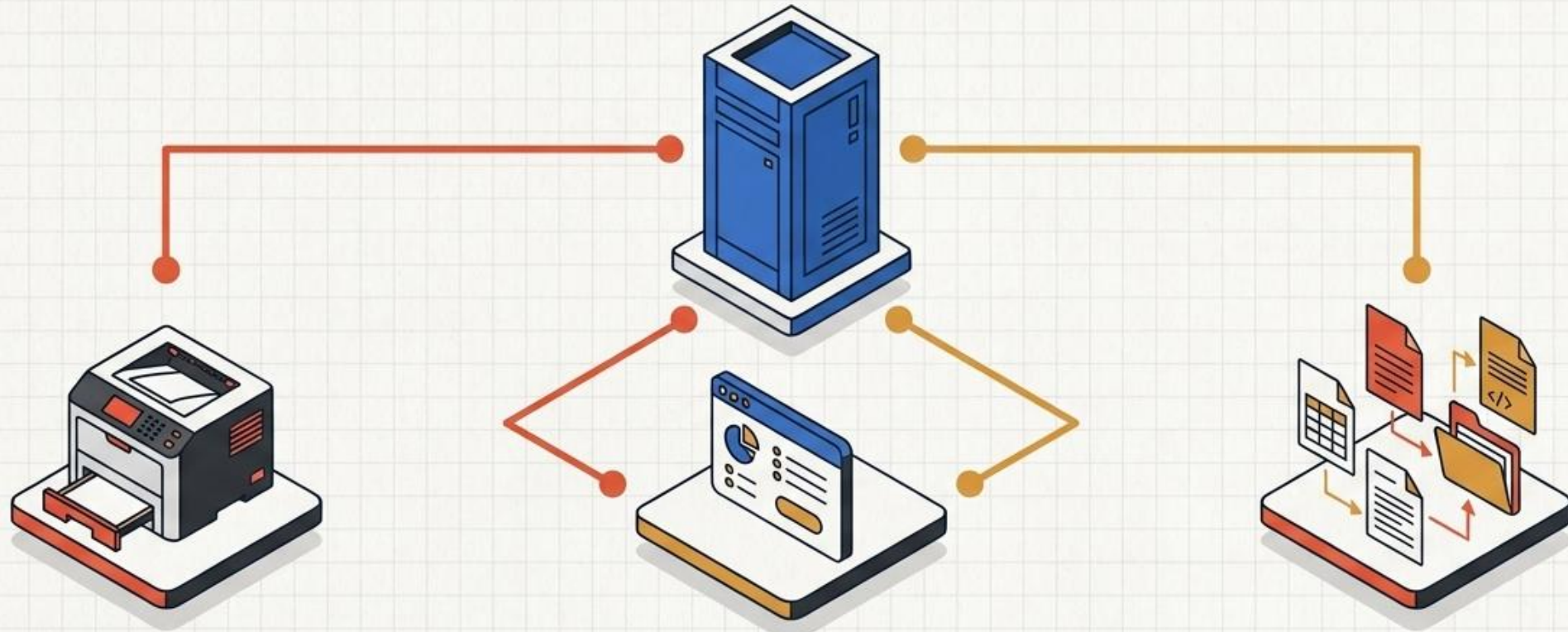
Why do we create a network?

1. Hardware resource sharing
2. Software resource sharing
3. File / data / information sharing
4. Fast communication

Networking saves time, saves cost, and makes collaboration easier.



Objectives of Networking



Hardware Resource Sharing

Multiple workstations accessing a single central printer or scanner.

Software Resource Sharing

Installing necessary software once on a central server rather than buying it for every individual device.

Data Resource Sharing

Accessing a centralized database from any connected computer.



Objective 1: Hardware Resource Sharing



Hardware means physical devices.

Examples:

- Printer
- Scanner
- Hard disk / storage
- Server
- Modem / router

Example: In an office, 20 computers can share one printer through a network.



Objectives 2–3: Software and Data Sharing



Software sharing:

- Same software can be stored on a server.
- Many users can use it through the network.

Data / file sharing:

- Assignments, marksheets, images, and databases can be shared.
- Users can access files with permission.



Objective 4: Communication



Computer networks help people communicate quickly.

Examples:

- Email
- Online class
- Video conference
- Instant messaging
- File transfer

During Google Meet class, which networking objectives are used?



Classification of Computer Networks



Computer networks can be classified in three major ways:

1. According to ownership

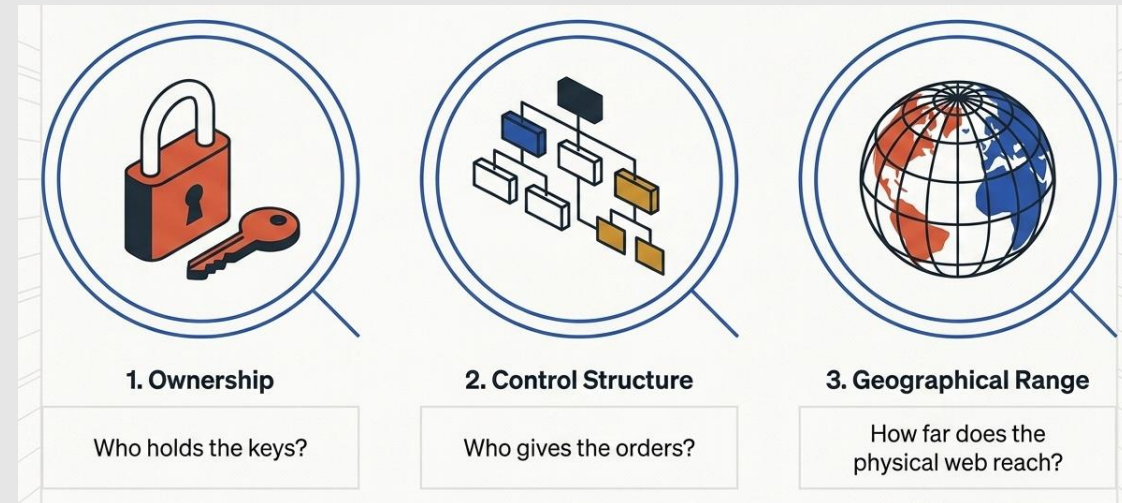
→ Private network, Public network

2. According to control structure

→ Peer-to-peer, Client-server, Hybrid

3. According to geographical range

→ PAN, LAN, CAN, MAN, WAN





Classification of Computer Networks



1. According to Ownership



Private Network

For private use only.



Public Network

Open access for the public.

2. According to Control Structure



Peer-to-Peer

Devices share resources directly.



Client-Server

Clients request, server provides.



Hybrid

Combination of peer-to-peer and client-server.

3. According to Geographical Range



PAN –

Personal Area Network



LAN –

Local Area Network



CAN –

Campus Area Network



MAN –

Metropolitan Area Network



WAN –

Wide Area Network

Smaller Coverage

Larger Coverage



By Ownership: Private and Public Network



Private Network

- Controlled by a person or organization.
- Only authorized users can access it.
- Security is usually higher.
- Example: school, bank, office network.

Public Network

- Many people can use it.
- Often managed by service providers.
- Security is comparatively lower.
- Example: mobile network, public Wi-Fi, Internet service.

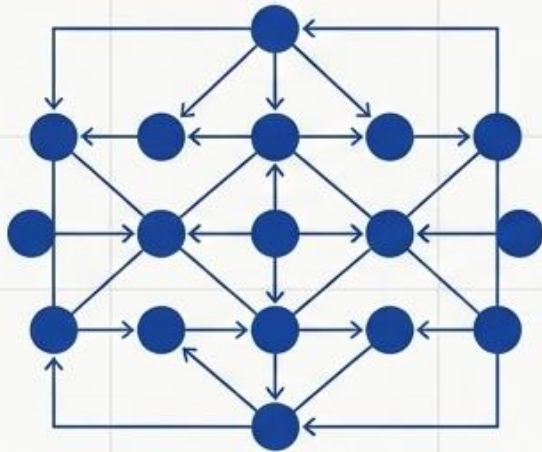


By Ownership: Private and Public Network



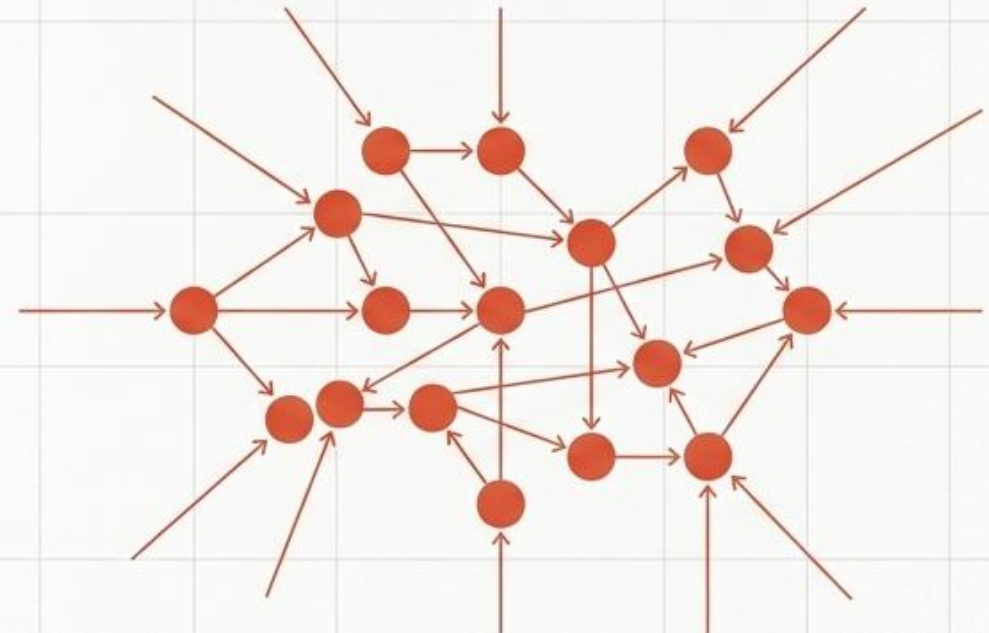
Private Network

Controlled by a single organization (e.g., a high school's internal network). Strong security, restricted access, high confidentiality.



Public Network

Managed by an organization but open for anyone to use, often for a fee (e.g., Mobile networks, Internet). High access, lower inherent security.





By Control Structure



Three common structures:

1. Peer-to-peer network

→ All computers are equal.

2. Client-server network

→ Server provides services to clients.

3. Hybrid network

→ Combination of peer-to-peer and client-server.



Peer-to-Peer Network

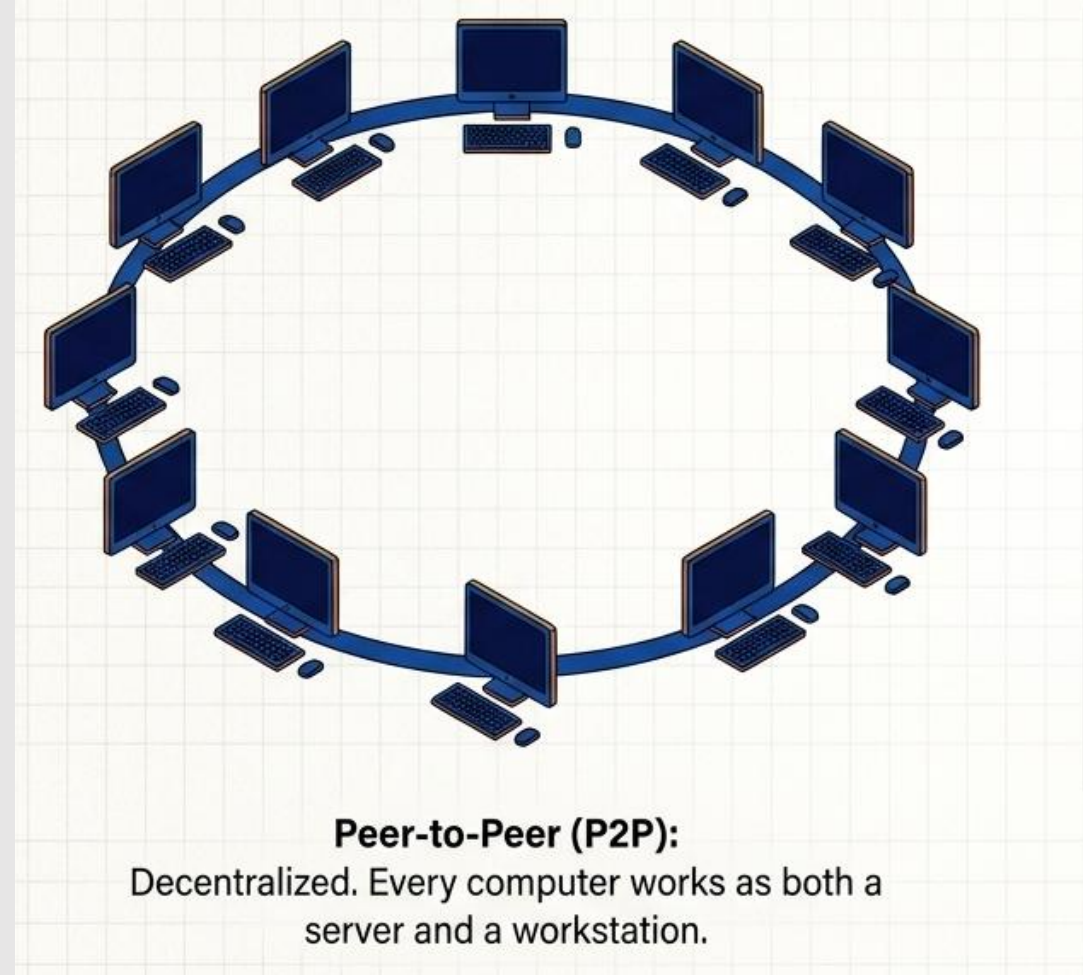


In a peer-to-peer network, every computer can share resources with every other computer. There is no dedicated server.

Features:

- All computers are equal.
- Low cost and easy to set up.
- Suitable for small homes or offices.
- Security is not controlled centrally.

Memory: Peer = Equal





Client-Server Network

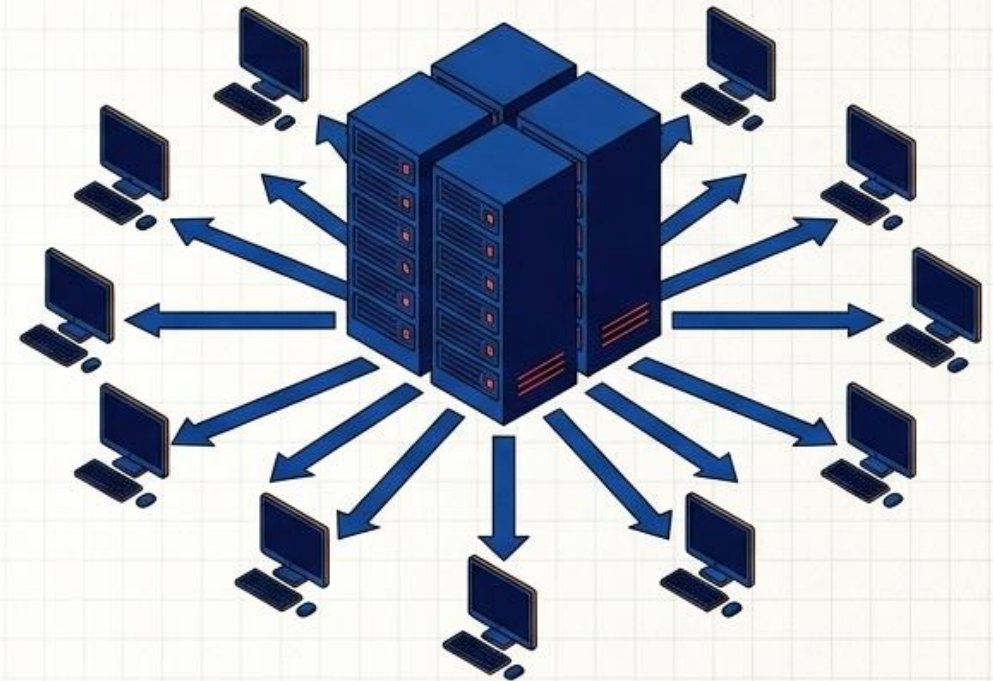


In a client-server network, clients request services and the server provides services.

Features:

- Central control by server.
- Better security and backup.
- Suitable for large organizations.
- Needs server hardware and administrator.

Memory: Client asks, Server gives.



Client-Server:

Centralized. A dedicated resource computer (server) provides services and security rules to other computers (clients).



Peer-to-Peer vs Client-Server



Feature	Peer-to-Peer	Client-Server
Control	Decentralized	Centralized
Server	No dedicated server	Dedicated server needed
Cost	Lower	Higher
Security	Lower / individual	Higher / centrally managed
Best for	Small home or office	Large organization



Hybrid Network



A hybrid network combines the features of peer-to-peer and client-server networks.

Example:

- Important software and files are controlled by the server.
- Users may also share some resources with each other.

Advantage:

- Flexible and useful for many organizations.

Disadvantage:

- More difficult to manage than a simple network.



By Geographical Range



Networks can also be classified by their coverage area.

PAN → Personal area

LAN → Local area

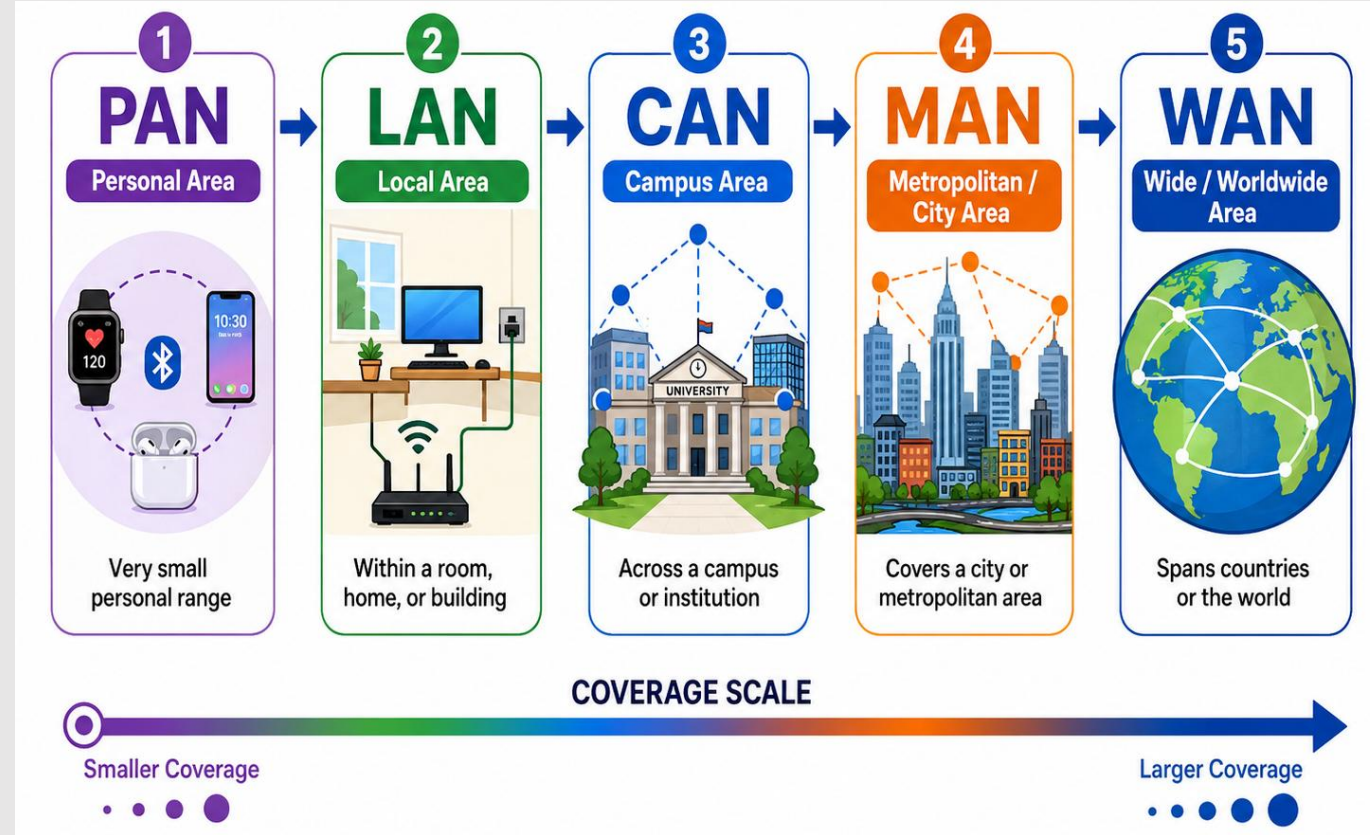
CAN → Campus area

MAN → Metropolitan / city area

WAN → Wide / worldwide area

Memory line:

Personal → Local → Campus → Metro
→ World





PAN — Personal Area Network



PAN is a very small network around one person.

Range:

- Usually up to 10 meters

Examples:

- Bluetooth headset
- Smartwatch connected to phone
- Wireless mouse / keyboard
- Phone connected to laptop

Memory: PAN = Personal



LAN — Local Area Network



LAN connects computers in a small local area.

Area:

- Room, building, office, school, or computer lab

Examples:

- School computer lab
- Office network
- Home Wi-Fi

Memory: LAN = Local



CAN and MAN



CAN — Campus Area Network

- Connects multiple LANs inside a campus.
- Example: university or college campus.

MAN — Metropolitan Area Network

- Connects networks within a city.
- Example: city cable TV or city government network.



WAN — Wide Area Network



WAN connects networks over a very large geographical area.

Area:

- City to city
- Country to country
- Continent to continent
- Worldwide

Examples:

- Internet
- Global company network
- International banking network

Memory: WAN = Worldwide



Geographical Spectrum

